

BEC305 — Python Programming

EC branch, semester 3, 2025 scheme. Phase 1 output, generated by raaghu-pc.

Python from the interactive shell to database applications — flow control and functions, the four built-in data structures, regular expressions and files, classes and objects, and web services with SQL.

What is in this folder

Path	Contents
python-programming.js	The machine-readable content file Phase 2 converts. 5 modules, 80 teaching units, quizzes and assignments.
ppt/	Five .pptx decks, 56 slides each, plus one .inspect.ndjson sidecar per deck.
quizzes/	Five quiz PDFs with answer keys, model answers and marking schemes, plus five Moodle GIFT files.
assignments/	Five assignment PDFs with full four-level rubrics and CO attainment tables.
pyq/	Copies of the branch-level merged question papers and PYQ_Index.pdf.
syllabus/	The syllabus extract, a copy of the source syllabus PDF, and Syllabus_Mapping.pdf.
MANIFEST.json	Counts, provenance and the result of every verification check.

Deck conventions Phase 2 depends on

- Every text shape carries a name: `title`, `objective` or `text`. There are no unnamed text shapes.
- No PowerPoint animation or slide transition XML is present.** Animation is written as an `Animation instructions:` line in the speaker notes, and Phase 2 rebuilds the motion in React from that line.
- Every slide has a notes block with all six headings in this order: `What to say:`, `Pause cue:`, `Pointing cue:`, `Question to ask students:`, `Animation instructions:`, `Transition style:`, followed by a `[Sources]` line.
- Each unit in the content file carries a `slides` array naming the deck slides that teach it. That array is the join key between the deck and the content file.
- Each unit carries a `visual` id and a `visualSpec`. The `visualSpec` reveal order and the matching slide's `Animation instructions:` line describe the same sequence.

Module index

Module	Title	Hours	Units	Slides	Chapters
1	Python Basics, Flow Control and Functions	9	16	56	TB1 (Sweigart 1e) Ch. 1 Python Basics, Ch. 2 Flow Control, Ch. 3 Functions, pp. 13-78
2	Data Structures: Lists, Dictionaries and Strings	8	16	56	TB1 (Sweigart 1e) Ch. 4 Lists, Ch. 5 Dictionaries and Structuring Data, Ch. 6 Manipulating Strings, pp. 79-144

Module	Title	Hours	Units	Slides	Chapters
3	Pattern Matching with Regular Expressions and File Handling	8	16	56	TB1 (Sweigart 1e) Ch. 7 Pattern Matching with Regular Expressions, Ch. 8 Reading and Writing Files, pp. 147-192
4	Classes and Objects	9	16	56	TB2 (Downey, Think Python 2e) Ch. 15-18 — TB2 was not supplied in this environment, so chapters are cited as the syllabus gives them and no page numbers are claimed
5	Networked Programs, Web Services and Databases	8	16	56	TB3 (Severance, Python for Everybody) Ch. 12, Ch. 13 up to 13.5, Ch. 15 — TB3 was not supplied in this environment, so chapters are cited as the syllabus gives them and no page numbers are claimed

Sources

- Syllabus: [3ecsy11.pdf](#), VTU EC semester 3, 2025 scheme.
- Prescribed textbook 1: Al Sweigart, *Automate the Boring Stuff with Python*, No Starch Press, 1st edition, 2015. File: [automatetheboringstuffwithpython_new.pdf](#). Machine-extractable: yes, roughly 1.8 million characters; all TB1 chapter and page references come from the book's own table of contents.
- Prescribed TB2 as listed by the syllabus: Allen B. Downey, *Think Python: How to Think Like a Computer Scientist*, Green Tea Press, 2nd edition, 2015. **Not supplied in this environment.** Used by the syllabus for the whole of Module 4 (Classes and objects, Chapters 15-18). Those units are aligned to the verbatim syllabus headings, cite only the chapter numbers the syllabus gives, and say so in their [Sources] note.
- Prescribed TB3 as listed by the syllabus: Charles R. Severance, *Python for Everybody: Exploring Data Using Python 3*, CreateSpace Independent Publishing Platform, 1st edition, 2016. **Not supplied in this environment.** Used by the syllabus for the whole of Module 5 (web services and databases, Chapters 12, 13 up to 13.5, and 15). Those units are aligned to the verbatim syllabus headings, cite only the chapter numbers the syllabus gives, and say so in their [Sources] note.
- Previous years' papers: [public/Previous Years QPs/EC/](#), copied unchanged. See [pyq/PYQ_Index.pdf](#) for the scheme-drift warning — Network Analysis is examined under BEC304 in those papers, not BEC305.

Regenerating

```
node temp/.tools/build-decks.mjs <subjectDir> python-programming.js
node temp/.tools/build-docs.mjs <subjectDir> python-programming.js <generatedBy> <pyqDir>
<syllabusPdf>
node temp/.tools/verify.mjs <subjectDir> python-programming.js
```

Generator scripts and `node_modules` live in `temp/.tools/`. Nothing outside `temp/` was written.